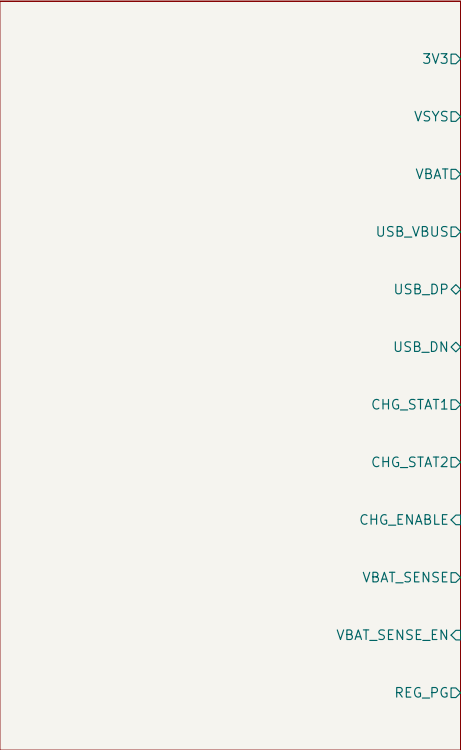


Cicala Rev A system block diagram

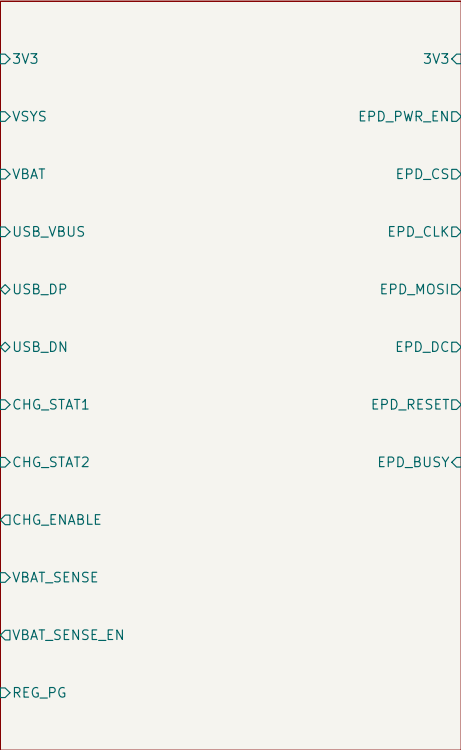
Power flow: USB-C 5 V or the protected cell -> BQ25185 power path (VSYSD)
-> TPS63802 buck-boost (3V3) -> TPS22917 load switches, one for the battery divider and one for the panel and its boost network.

Power and USB



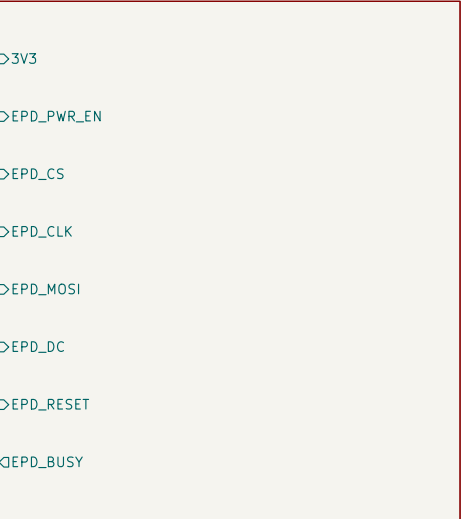
File: sheets/battery_charger.kicad_sch

Controller and user I/O



File: sheets/esp32_core.kicad_sch

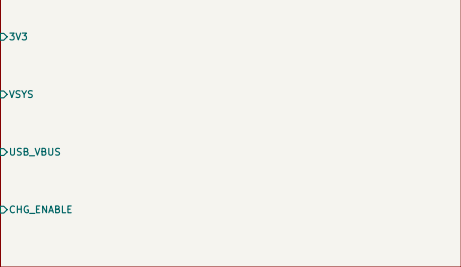
E-paper display



File: sheets/epaper.kicad_sch

GND is a global net and is not drawn here.

Charge temperature guard



File: sheets/charge_temperature.kicad_sch

Cicala

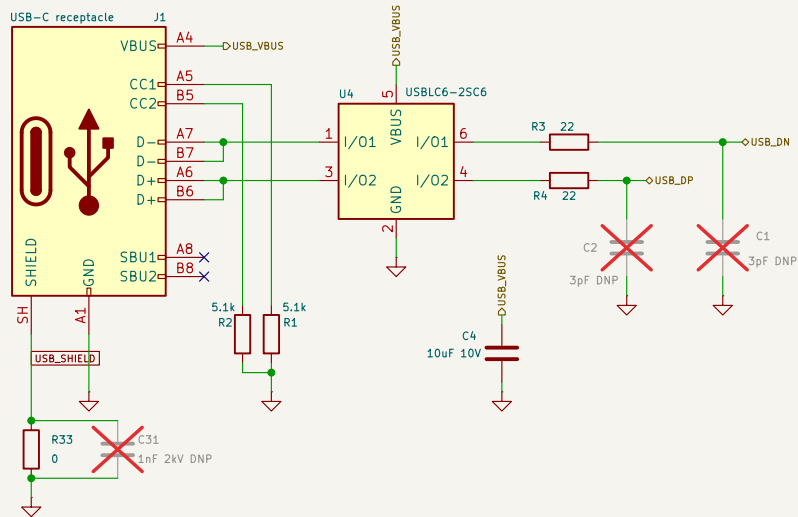
Sheet: /
File: cicala_rev_a.kicad_sch

Title: Cicala Rev A schematic overview

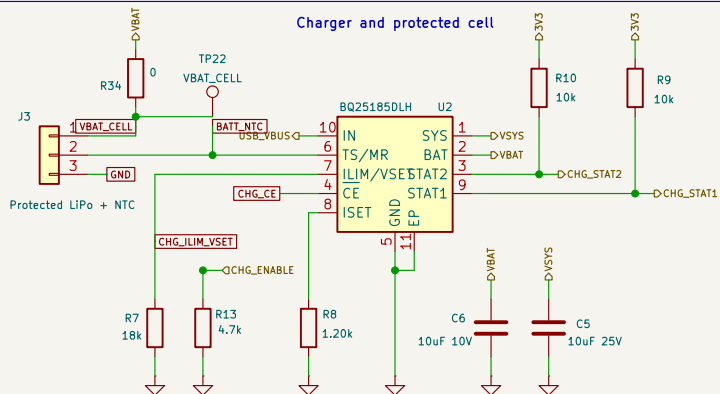
Size: A4	Date: 2026-09-07	Rev: Rev A
KiCad E.D.A. 10.0.5		Id: 1/5

D

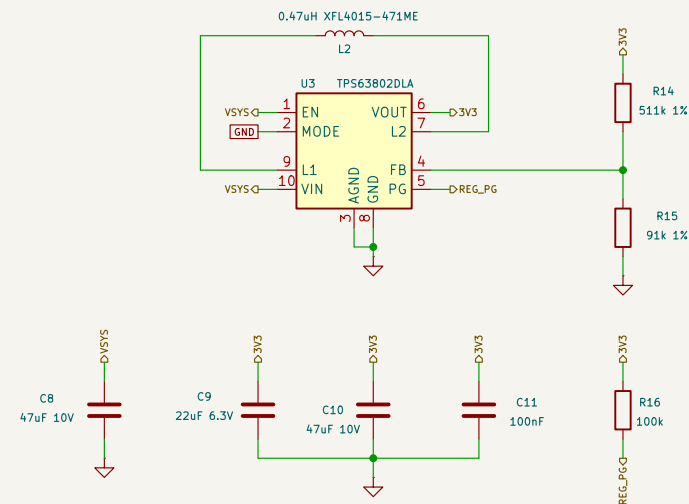
USB-C receptacle J1



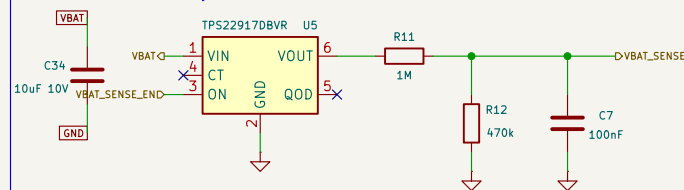
Charger and protected cell



3.3 V buck–boost

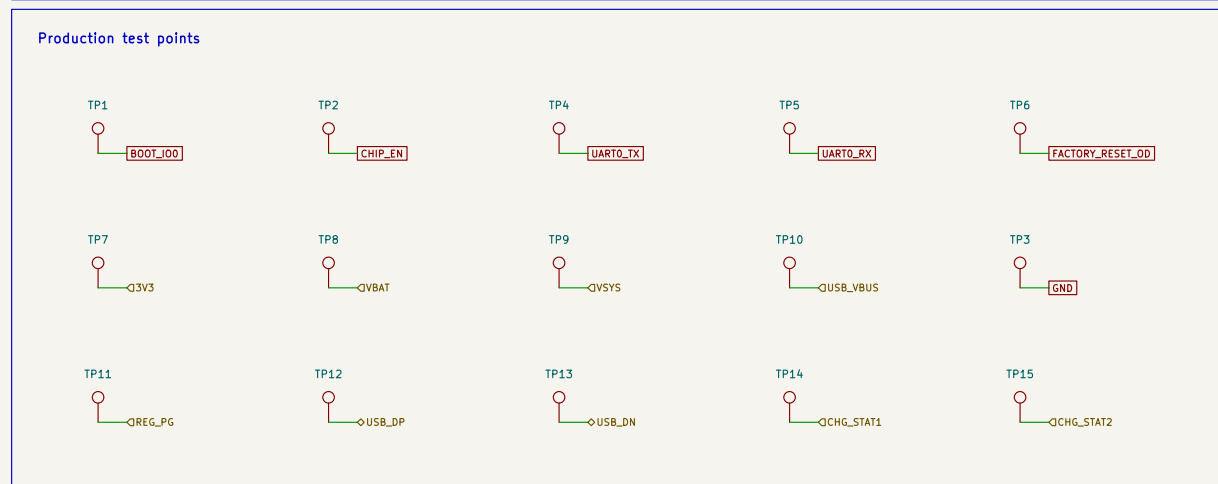
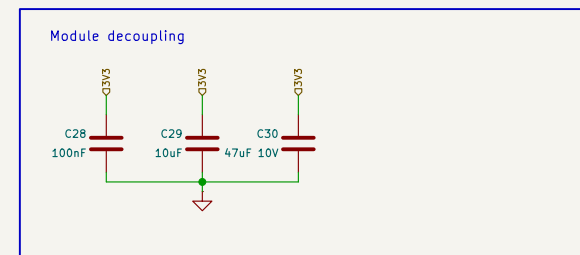
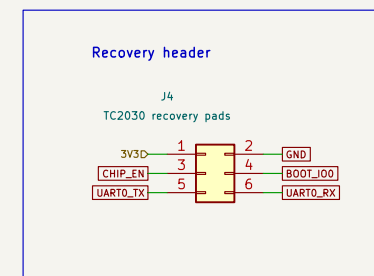
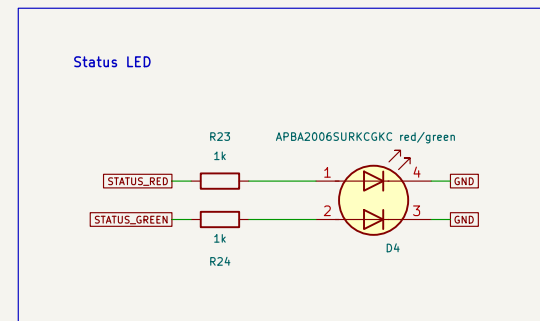
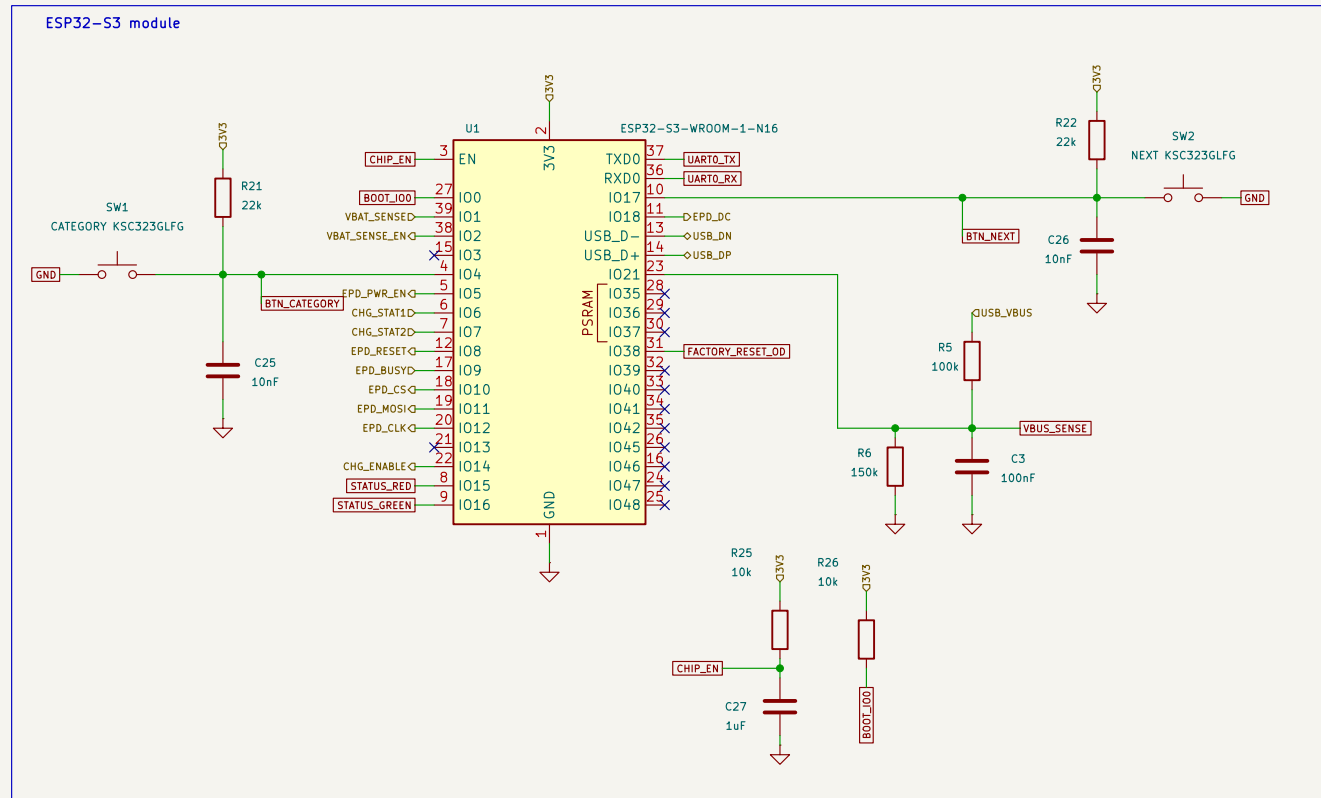


Switched battery measurement



Id: 2/5

WROOM-1 is preferred; WROOM-1U remains the placement/RF fallback. USB is the normal debug path.



Cicala

Sheet: /Controller and user I/O/

File: esp32_core.kicad_sch

Title: Controller, user I/O, and recovery

Size: A4	Date: 2026-09-07
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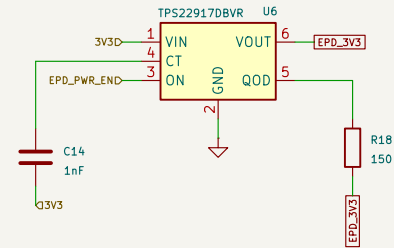
Rev: Rev A

Size: A4	
KiCad E.D.A. 10.0.5	

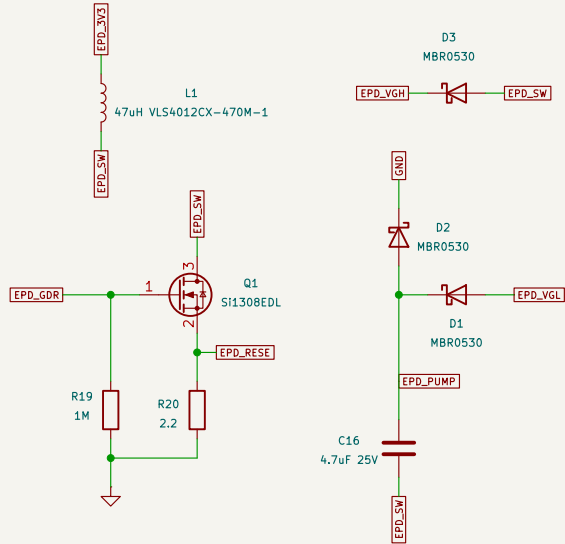
Id: 3/5

The load switch removes panel and boost power between refreshes.
Boost network follows Good Display's reference.

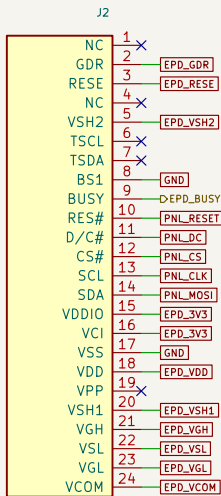
Switched panel supply



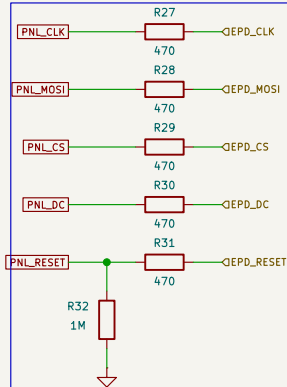
SSD1680 boost network



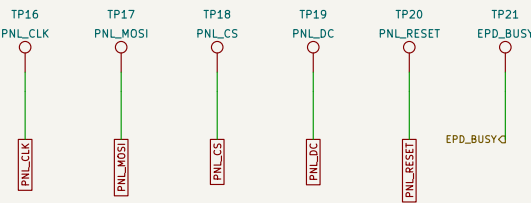
Panel FPC



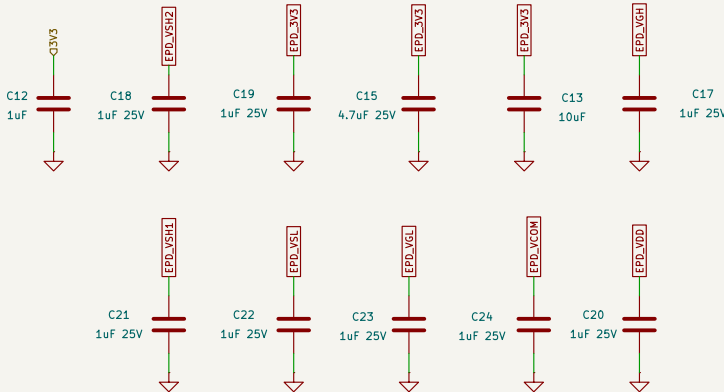
Bus isolation



Panel bus test points



Panel reservoir capacitors



Cicala

Sheet: /E-paper display/
File: epaper.kicad_sch

Title: GDEY0213B74 e-paper interface and boost

Size: A4

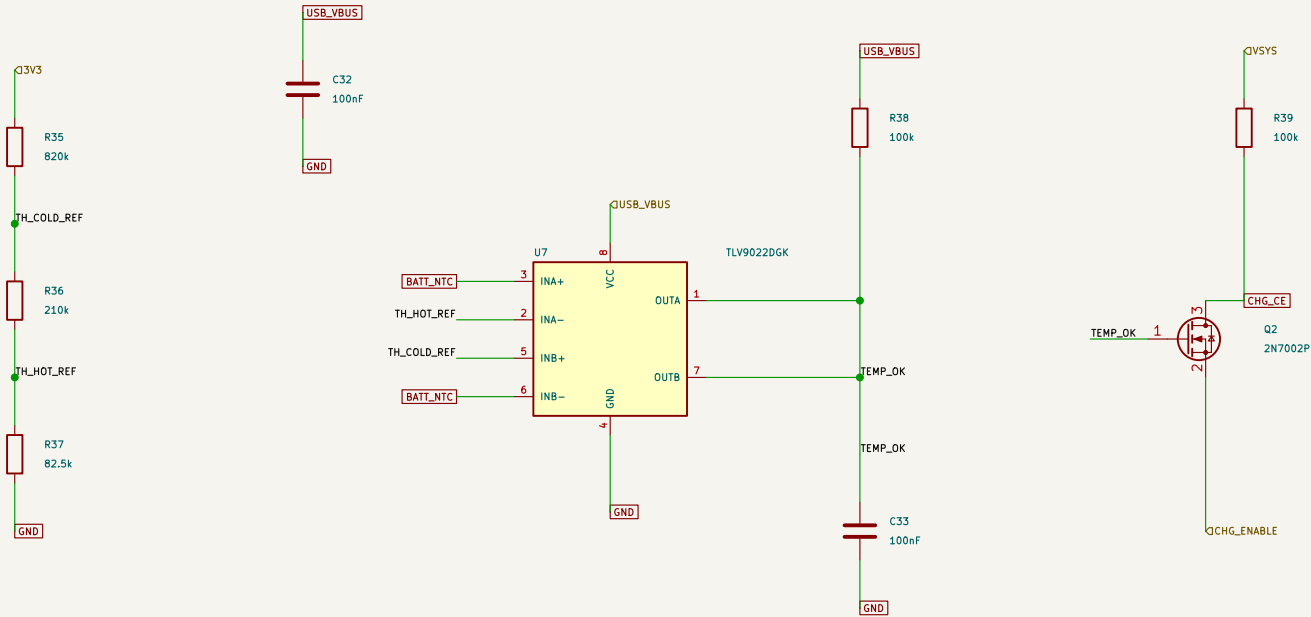
Date: 2026-09-07

KiCad E.D.A. 10.0.5

Rev: Rev A

Id: 4/5

Independent battery temperature veto



J3.2 / BATT_NTC remains connected to BQ25185 TS/MR.
Semitec 103AT-2 is bonded to the protected cell and returns to J3.3.

Both comparator outputs must release TEMP_OK to permit charging.
An open or shorted thermistor disables charging. C33 delays permission at USB insertion.
Firmware CHG_ENABLE high disables charging; R13 supplies the low default through Q2.

3V3 sets the comparison thresholds; U7 and the gate pull-up run only from USB.
The 1.1125 Mohm reference ladder draws 2.97 uA from 3V3.

Cicala		
Sheet: /Charge temperature guard/ File: charge_temperature.kicad_sch		
Title: Hardware charge temperature guard		
Size: A4	Date: 2026-09-07	Rev: Rev A
KiCad E.D.A. 10.0.5		Id: 5/5